

Abis Production Photo & Video Studio

Complete Web Application Functionality & Architectural Specification

Author: Antigravity Engineering | Platform: Next.js 15, React 19, Supabase, Tailwind CSS, Three.js | Version: 1.0

Executive Overview: Abis Production is a state-of-the-art web platform engineered for East Africa's premier luxury photography and videography studio. It combines a visually captivating public studio interface, an interactive 7-step booking engine, real-time Ethiopian payment tracking (Telebirr, CBE Birr, Bank Transfer), a client gallery proofing portal, and a full-suite Admin Control Panel (CRM/ERP) with hybrid Supabase cloud and IndexedDB offline persistence.

1. Public Website & Client Experience

1.1 Luxury Homepage (/)

- Dynamic Hero Section:** Features 3D Canvas visual effects, brand taglines, and instant call-to-action buttons ('Book Session' and 'Explore Portfolio').
- Live Portfolio Showcase:** Interactive photo gallery with real-time category filtering (Wedding, Graduation, Birthday, Family, Studio, Corporate, Outdoor, Engagement, Christening).
- Featured Services & Tiered Pricing:** Direct service cards detailing package highlights, duration, and starting prices in Ethiopian Birr (ETB).
- Interactive Client Testimonials:** Verified client quotes and reviews highlighting photography quality and studio professionalism.
- Brand Heritage & Footer:** Complete studio information including office address (Bole Atlas, Addis Ababa), direct hotline, phone, email, and social links (Instagram, TikTok, Facebook, WhatsApp).

1.2 Interactive 7-Step Online Booking Engine (/booking)

The platform provides a guided step-by-step booking workflow designed for effortless client onboarding:

Step	Workflow Phase	Functionality & User Capabilities
Step 1	Service Category	Select from 10 specialized categories (Wedding, Birthday, Christening, Engagement, Outdoor, Graduation, Corporate, Studio, Family, Video Production).
Step 2	Package Selection	Choose tier options (Standard, Gold Royal, Platinum Cinema, VIP) with instant price calculations in ETB.
Step 3	Date Picker Engine	Interactive date selector validated against studio-blocked dates and existing bookings to prevent double-booking.
Step 4	Time Slot Picker	Select preferred session times (08:00 AM, 10:00 AM, 12:00 PM, 02:00 PM, 04:00 PM, 06:00 PM).
Step 5	Customer Details	Enter customer name, phone, email, event type, venue location/hotel address, and special instructions.
Step 6	Booking Summary	Review complete order summary, date/time breakdown, venue details, and calculated grand total.
Step 7	Payment & Proof Upload	Select Ethiopian payment method (Telebirr, CBE Birr, Bank Transfer, Cash Deposit), copy account details for transfer, and upload event proof photos.

1.3 Order Tracking & Client Photo Proofing Portal (/track-order)

- **Booking Lookup:** Clients enter their Reference Number (e.g. PS-2026-884920) and Email Address to track real-time booking status.
- **Live Status Timeline:** Visual progress pipeline displaying current status: Pending Review → Payment Verified → Booking Confirmed → Photos Uploaded → Selection Submitted → Completed.
- **Receipt Proof Verification Inspector:** Preview uploaded payment receipt slip image or PDF document and verify payment approval status.
- **Client Gallery Access via Secret 6-Digit PIN:** Secure client portal protection for viewing private high-res event photos.
- **Photo Selection & Favoriting Engine:** Clients select their favorite photos for master editing up to the allowed limit, leave comments on individual photos, and submit final selections directly to the studio editing team.

1.4 Portfolio Showcase & BTS Video Player (/portfolio)

- Filterable high-definition image showcase grouped by event types.
- Full-screen Lightbox with metadata, client names, and event dates.
- Integrated 4K Behind-the-Scenes (BTS) Video Player modal for cinematic video previews.

2. Administration Control Panel & CRM/ERP Hub (/admin)

The Admin Control Panel equips studio managers with full operational oversight across 11 integrated modules:

Module Name	Operational Capabilities & Studio Management Features
Dashboard Overview	Real-time KPI metrics: Total Verified Revenue (ETB), Pending Payment Proof Counter, Total Bookings Count, and
Bookings Management	Search, filter, and review all customer bookings. Verify or reject payment receipts, inspect uploaded receipt image
Client Gallery Delivery	Upload high-res photo packages to client galleries, assign secret PINs, configure max photo selection limits, and
Portfolio CMS	Add, edit, delete, and reorder showcase items. Batch image uploader with automatic client-side canvas compress
Services Manager	Manage service titles, categories, starting prices, estimated durations, cover photos, and toggle active/disabled a
Packages Manager	Configure package tiers per service, tagline descriptions, prices in ETB, duration, photographer counts, feature b
Customers CRM	Customer relationship records detailing full name, phone number, email address, total sessions booked, and lifet
Staff Access Control	Role-Based Access Control (RBAC) supporting Super Admin, Admin, and Editor roles. Create new admin creden
Payment Settings	Configure Telebirr, CBE Birr, Bank Transfer, and Cash Deposit account names, numbers, and payment instructio
Calendar & Blocked Dates	Manage studio booking availability calendar. Block target dates with custom reasons (maintenance, holidays, priv
Website Content CMS	Live editable site settings for homepage hero title, subtitle, hero image, about section content, office address, pho

3. Technical Architecture & Database Integration

3.1 Hybrid Persistence Layer (Cloud & Offline Resilience)

- **Cloud Database (Supabase PostgreSQL):** Primary data store for production synchronization across public clients and admin controllers. Tables include bookings, services, packages, blocked_dates, portfolio_items, payment_settings, site_settings, and admin_users.
- **Local Offline Persistence (IndexedDB):** Client-side storage engine using idbStore.ts to cache data locally. Ensures zero-downtime performance and offline data retention if internet access is interrupted.

3.2 Image Optimization & Performance Engine

- **Client-Side Canvas Compression:** High-resolution client and portfolio image uploads are automatically compressed via HTML5 Canvas (1920x1920 @ 82% WebP/JPEG quality) before storage to minimize bandwidth consumption and guarantee ultra-fast page loads.
- **Real-Time Data Sync:** Event listeners automatically refetch data when window focus shifts or DB mutations occur, ensuring payment settings, pricing changes, and calendar blocks reflect instantly on the public website.

Summary of System Capabilities

Conclusion: The Abis Production Photo & Video Studio web application delivers an end-to-end luxury photography business solution. From high-converting public portfolios and dynamic 7-step bookings to real-time Ethiopian payment tracking, secure client photo delivery, and comprehensive CRM/ERP administration, the platform operates seamlessly across local and cloud environments.